

Signals and Systems

(EE2008)

Date: March 22nd,
2025

Course Instructor(s)

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Roll No. _____ Section _____
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Student Signature _____

Sessional-II Exam

Solution

Total Time (Hrs.): 1

Total Marks: 30

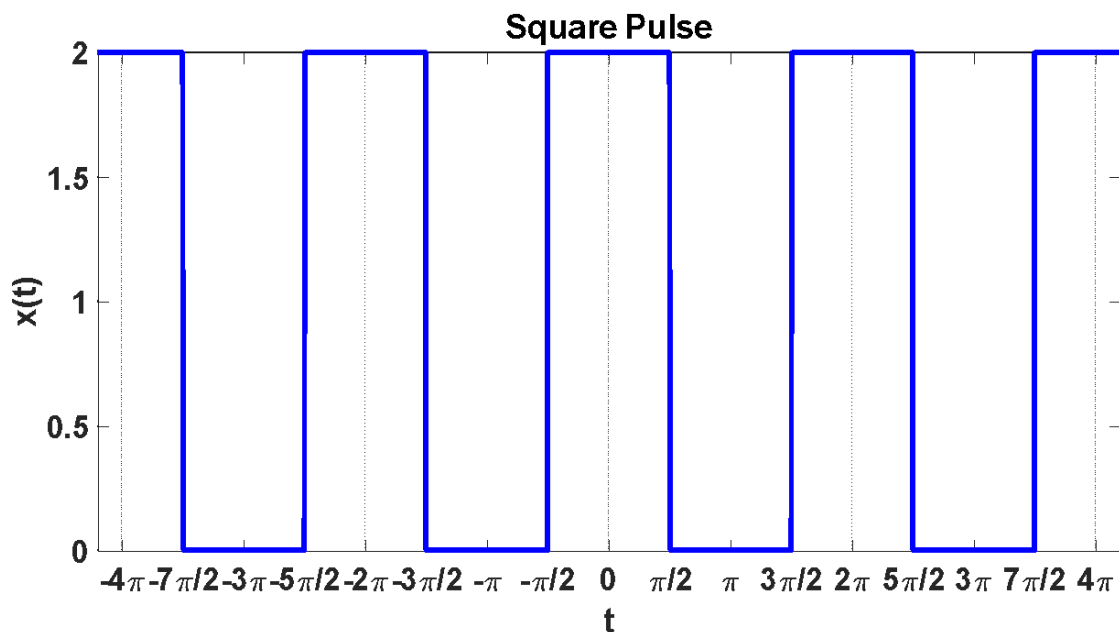
Total Questions: 3

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Attempt all the questions.

CLO 3: Analyze signals using Fourier series representation.

Q1: **Analyze** the coefficients of the compact trigonometric Fourier series for the given square-pulse periodic signal shown below and sketch its amplitude and phase spectra. [10 marks]



SOLUTION:

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos nt + b_n \sin nt.$$

$$\text{Ans } \omega_0 T_0 = 2\pi$$

$$\omega_0 (2\pi) = 2\pi$$

$$\boxed{\omega_0 = 1}$$

$$a_0 = \frac{1}{T_0} \int_{T_0} x(t) dt$$

$$a_0 = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} 2 dt.$$

$$a_0 = \frac{2}{2\pi} \left[t \right]_{-\pi/2}^{\pi/2} \Rightarrow \frac{1}{\pi} \left[\pi/2 - (-\pi/2) \right] \Rightarrow \frac{1}{\pi} [\pi] = 1$$

$$a_n = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} 2 \cos nt dt \Rightarrow \frac{2}{\pi} \int_{-\pi/2}^{\pi/2} \cos nt dt \Rightarrow \frac{2}{\pi} \left[\frac{1}{n} \sin nt \right]_{-\pi/2}^{\pi/2}$$

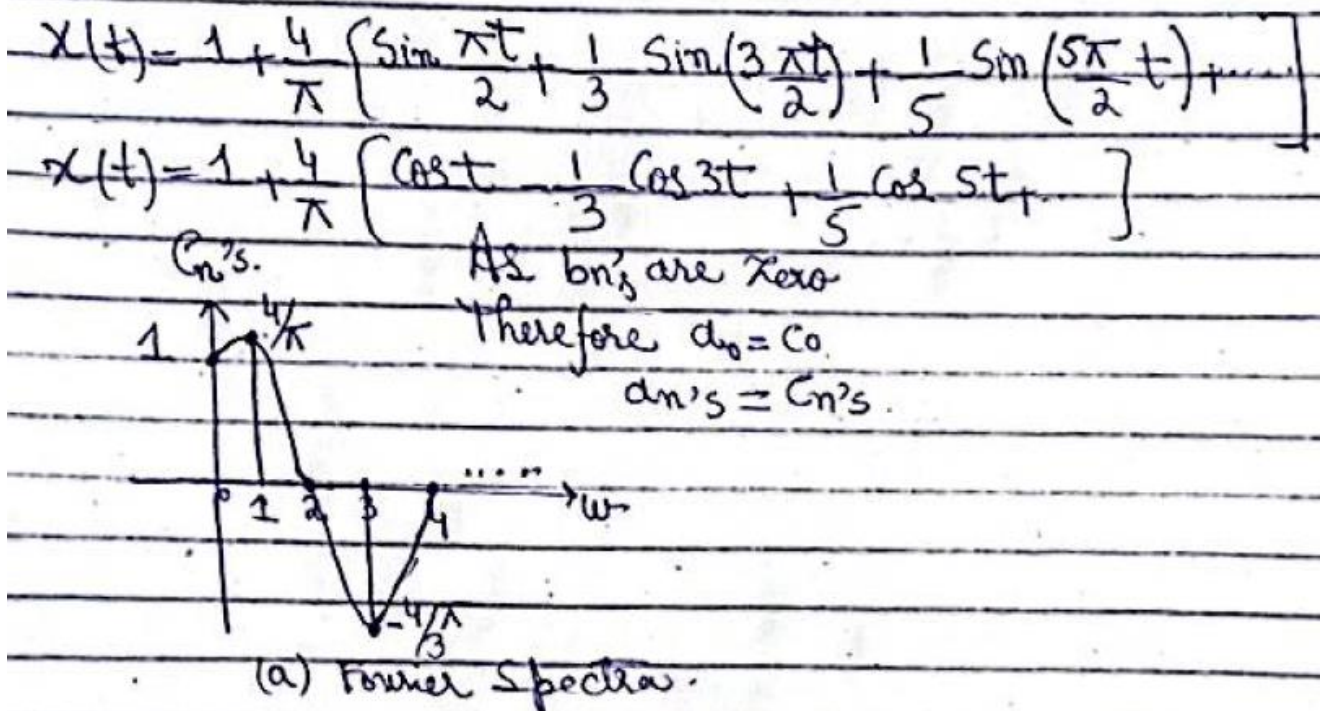
$$a_n = \frac{2}{n\pi} \left[\sin n\pi/2 - \sin n(-\pi/2) \right] \Rightarrow \frac{2}{n\pi} \left[\sin(n\pi/2) + \sin(n\pi/2) \right]$$

$$a_n = \frac{2}{n\pi} \left[2 \sin(n\pi/2) \right] \Rightarrow \frac{4}{n\pi} \left[\sin(n\pi/2) \right].$$

$$a_n = \begin{cases} 0; & \text{even} \\ \frac{4}{n\pi}; & n=1, 5, 9, 13, \dots \\ -\frac{4}{n\pi}; & n=3, 7, 11, 15, \dots \end{cases}$$

$$b_n = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} 2 \sin nt dt \Rightarrow \frac{2}{\pi} \int_{-\pi/2}^{\pi/2} \sin nt dt \Rightarrow \frac{2}{\pi} \left[-\frac{1}{n} \cos nt \right]_{-\pi/2}^{\pi/2}$$

$$b_n = \frac{2}{\pi} \left[\frac{1}{n} \left(\cos n\pi/2 - \cos n(-\pi/2) \right) \right] = \frac{2}{\pi} [0] \Rightarrow 0.$$



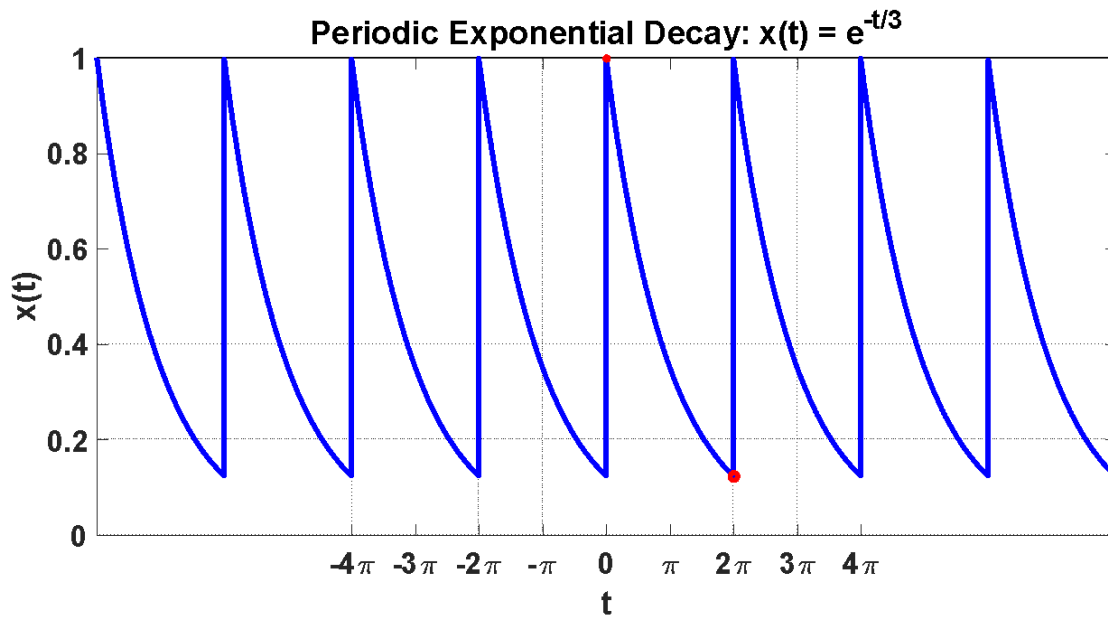
$$\theta_n = \tan^{-1} \left(-\frac{b_n}{a_n} \right) \Rightarrow \tan^{-1} \left(\frac{0}{4/n\pi} \right) = 0.$$

$$\theta_n = \begin{cases} 0 & \text{for all } n \end{cases}$$

(6.17). This means that phases of all components are zero, and we can discard the phase spectrum and manage with only the amplitude spectrum, as shown in Fig. 6.5b. Observe that there is no loss of information in doing so and that the amplitude spectrum in Fig. 6.5b has the complete information about the Fourier series in Eq. (6.17). Therefore, whenever all sine terms vanish ($b_n = 0$), it is convenient to allow C_n to take on negative values. This permits the spectral information to be conveyed by a single spectrum.^[†]

CLO 3: Analyze signals using Fourier series representation.

Q2: **Analyze** the coefficients of the exponential Fourier series for the given exponential signal shown below. [10 marks]



SOLUTION:

$$\omega_0 T_0 = 2\pi$$

$$\omega_0 2\pi = 2\pi$$

$$\omega_0 = 1$$

where

$$D_n = \frac{1}{T_0} \int_0^{2\pi} e^{-t/3} e^{-jnt} dt$$

$$D_n = \frac{1}{2\pi} \int_0^{2\pi} e^{-(\frac{1}{3} + jn)t} dt$$

$$D_n = \frac{1}{2\pi} \left[-\frac{1}{(\frac{1}{3} + jn)} e^{-(\frac{1}{3} + jn)t} \right]_0^{2\pi}$$

$$D_n = \frac{1}{2\pi} \left[-\frac{1}{(\frac{1}{3} + jn)} \left(e^{-(\frac{1}{3} + jn)2\pi} - e^{-(\frac{1}{3} + jn)0} \right) \right]$$

$$D_n = \frac{1}{2\pi} \left[-\frac{1}{\frac{1}{3} + jn} \left(e^{-2\pi/3 - j2n\pi} - e^0 \right) \right]$$

$$D_n = \frac{1}{2\pi} \left[-\frac{1}{\frac{1}{3} + jn} (0.123(1) - 1) \right]$$

$$D_n = \frac{1}{2\pi} \left[-\frac{1}{\frac{1}{3} + jn} (-0.876) \right]$$

$$D_n = \frac{1}{2\pi} \left[\frac{0.88}{\frac{1}{3} + jn} \right]$$

$$D_n = \frac{1}{6.28} \left[\frac{0.88}{\frac{1}{3} + jn} \right] \Rightarrow \frac{5.8}{\frac{1}{3} + jn} \Rightarrow \frac{5.8}{1 + j3n}$$

$$D_n = 0.16 \left(\frac{0.88}{\frac{1}{3} + jn} \right) \Rightarrow (0.16) \left(\frac{0.88 \cdot 3}{1 + j3n} \right) \Rightarrow \frac{0.42}{1 + j3n}$$

and

$$x(t) = \sum_{n=-\infty}^{\infty} \frac{1(0.42)}{1 + j3n} e^{jnt}$$

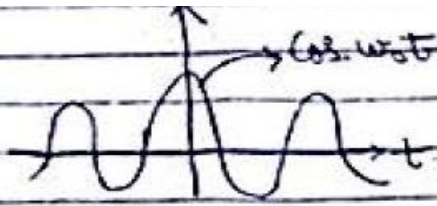
$$x(t) = 0.42 \sum_{n=-\infty}^{\infty} \frac{e^{jnt}}{1 + j3n}$$

$$x(t) = 0.42 \left[1 + \frac{1}{1 + j3} e^{jt} + \frac{1}{1 + j6} e^{j2t} + \dots \right]$$

CLO 4: Analyze signals using Fourier transform and its properties.

Q3: Analyze the Fourier transform of the everlasting sinusoid $\cos \omega_0 t$. Also plot $\cos \omega_0 t$ in the time domain. [5 marks]

Solution:



$$\mathcal{F}\{x(t)\} = \mathcal{F}\{\cos w_0 t\} \rightarrow X(\omega)$$

$$X(\omega) = \mathcal{F}\{\cos w_0 t\} \rightarrow \mathcal{F}\left[\frac{e^{jw_0 t} + e^{-jw_0 t}}{2}\right]$$

We know that

$$\mathcal{F}\{e^{jw_0 t}\} = 2\pi \delta(\omega - w_0)$$

And

$$\mathcal{F}\{e^{-jw_0 t}\} = 2\pi \delta(\omega + w_0)$$

Hence,

$$\mathcal{F}\{\cos w_0 t\} = \frac{1}{2} \left[\mathcal{F}\{e^{jw_0 t}\} + \mathcal{F}\{e^{-jw_0 t}\} \right]$$

$$= \frac{1}{2} \left[2\pi \delta(\omega - w_0) + 2\pi \delta(\omega + w_0) \right]$$

$$= \frac{2}{2} \left[\pi \delta(\omega - w_0) + \pi \delta(\omega + w_0) \right]$$

$$\mathcal{F}\{\cos w_0 t\} = \pi \left[\delta(\omega + w_0) + \delta(\omega - w_0) \right]$$

$\xrightarrow{\text{Impulse at } -w_0}$ $\xrightarrow{\text{Impulse at } w_0}$
 $\omega = -w_0$ $\omega = w_0$

